

Mapping the Missing Market Pathways in U.S. Dairy Regions

We introduce the *Dairy Market Pathways Index*, a new county-level measure combining direct sales, intermediated sales, and nearby dairy-sourcing food hubs. Higher values indicate a stronger measured local-market environment surrounding dairy production, addressing a key challenge: abundant milk production does not necessarily ensure access to these markets. A preliminary national assessment led us to compare the Upper Midwest (Michigan, Minnesota, and Wisconsin) and the Northeast (New York, Pennsylvania, and Vermont), two major dairy regions with contrasting market and processing structures. These pathways matter both to producers seeking additional buyers and value-added opportunities and to institutions, including schools, seeking dependable local suppliers. Although the index does not isolate farm-to-school sales, fluid milk accounts for more than half of local school food spending in both regions. Northeast school food authorities were also more likely to purchase local fluid milk (63% versus 48%) and other local dairy products (45% versus 23%)¹. These differences illustrate how access to local and regional markets depends not only on milk production but also on suitable processing, storage, and delivery channels.

Visualization 1 draws on FAME 2.0, the 2022 Census of Agriculture, and the USDA Local Food Portal. We first use the Census of Agriculture to identify counties with at least 100 milk cows. For each county in this sample, the index combines three measures: the share of all farm sales made directly to consumers through channels such as farmers markets, farm stands, and community-supported agriculture; the share made through intermediated local channels such as restaurants, grocery stores, schools, hospitals, and regional food aggregators; and the number of dairy-sourcing food hubs within 75 miles. Using sales shares rather than dollar values allows us to compare the market orientation of counties with farm economies of different sizes. We standardize the three components across this national sample of dairy counties and take an equally weighted average. Visualization 1 shows generally stronger measured pathways in Northeast dairy counties than in the Upper Midwest, along with substantial variation within each region.

Visualization 2 shows where milk reported through Federal Milk Marketing Orders (FMMO) originated and the types of regulated dairy plants operating in each region.² The mapped counties reported about 1.9 billion pounds of milk in the Upper Midwest and 2.2 billion in the Northeast, but that milk enters different processing systems. The Upper Midwest has 40 manufacturing-oriented plants and 16 fluid-bottling plants, compared with 8 and 34, respectively, in the Northeast. Of every 100 pounds reported through the Upper Midwest order, about 83 were used for cheese and whey. Northeast order milk was divided more evenly among fluid milk, soft products, cheese and whey, and butter and powder. The data cannot connect an individual farm to a specific plant or indicate how large each plant is. They do show that the two regions report similar amounts of milk but differ in the types of plants they contain and how milk is classified by use. Visualization 2 shows that similarly large milk-producing regions can be organized around different processing systems. The Upper Midwest is more specialized in manufacturing milk into cheese and whey, while the Northeast combines more fluid-bottling plants with a more balanced dairy product mix.

The visualizations identify places where strong dairy production coexists with weaker measured local-market pathways, as well as places where stronger pathways surround a smaller dairy base. The *Dairy Market Pathways Index* provides a county-level screening tool that USDA AMS can combine with production, infrastructure, and awards data to identify areas for further research or targeted support. Dairy-related Resilient Food Systems Infrastructure awards³ demonstrate one application. Upper Midwest awards emphasized cheese processing and related packaging, storage, and transportation, while Northeast awards covered a broader product mix and included cold storage, distribution, aggregation, and institutional-market access. Comparing award locations with index scores could show whether investments are reaching dairy-producing counties with weaker pathways and which supply-chain gaps remain. Producers, extension professionals, and regional organizations can use the scores and components to compare local conditions and identify missing connections, while researchers can update the index or adapt it to other agricultural sectors.

¹USDA Food and Nutrition Service, *2023 Farm to School Census National and State Data Workbook*. Authors' calculation.

²FMMOs cover about 75% of U.S. milk production, providing a broad though incomplete view of the regulated milk supply.

³USDA Agricultural Marketing Service, "Resilient Food Systems Infrastructure Program,"